

CSA1718-ARTIFICIAL INTELLIGENCE

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1. Water Jug Problem

Method: Depth-First Search (DFS)

Problem Understanding

Represent every jug configuration as a state and search deeply until the goal is found.

Method Selection

Depth-First Search (DFS) explores one path completely before backtracking.

Solution Steps

1. Start with both jugs empty **(0,0)**.
2. Fill or empty jugs and pour water between them.
3. Continue exploring one sequence of states.
4. If the goal is not reached, backtrack and try another sequence.
5. Stop when the state **(2,0)** is reached.

Result

The 4-gallon jug contains exactly **2 gallons**.

Interpretation

DFS is simple to implement and explores one solution path at a time until the goal state is found.

2. Mars Rover

Method: Goal-Based Agent

Problem Understanding

The rover's main goal is to safely complete its mission.

Method Selection

A **Goal-Based Agent** chooses actions based on achieving predefined goals.

Percepts

- Camera images
- Rock samples
- Temperature
- Obstacles
- Battery level
- Terrain information

Operating Environment

- Partially Observable
- Dynamic
- Sequential
- Single Agent
- Continuous

Actions

- Move
- Turn
- Capture images
- Drill rocks
- Collect samples
- Send data
- Recharge

Performance Measure

- Safe navigation
- Maximum sample collection
- Low energy usage
- Successful data transmission

Interpretation

A Goal-Based Agent focuses on achieving mission objectives while adapting its actions as needed.

3. 8-Queens Problem

Method: Constraint Satisfaction Problem (CSP)

Problem Understanding

Place 8 queens so that no two queens attack each other.

Problem Components

Variables

- Q1, Q2, Q3, ..., Q8

Domain

- Rows 1 to 8

Constraints

- No two queens in the same row.
- No two queens in the same column.
- No two queens on the same diagonal.

Method Selection

Treat the problem as a **Constraint Satisfaction Problem (CSP)**.

Solution Process

1. Assign a row to each queen.
2. Check all constraints.
3. If a constraint is violated, choose another row.
4. Continue until all constraints are satisfied.

Interpretation

CSP reduces unnecessary search by enforcing constraints while assigning values.

4. OLA Cab Booking

Method: Simple Reflex Agent

Problem Understanding

The customer selects a cab, and the application responds based on the current situation.

Method Selection

A **Simple Reflex Agent** follows predefined rules without considering past states.

Pseudocode

START

Input pickup location

Input destination

Display available cabs

IF customer selects Mini

 Book Mini

ELSE IF customer selects Sedan

 Book Sedan

ELSE IF customer selects Prime

 Book Prime

ELSE

 Show "No Cab Available"

Display Driver Details

END

Interpretation

The Simple Reflex Agent makes decisions based only on the current input, making it suitable for straightforward booking tasks.

5. Logistics Company

Method: Greedy Best-First Search (GBFS)

Problem Understanding

Find a path from warehouse **S** to warehouse **G**.

Method Selection

Greedy Best-First Search always expands the node estimated to be closest to the destination.

Solution Steps

1. Start at warehouse **S**.
2. Evaluate neighbouring warehouses using heuristic values.
3. Select the warehouse with the smallest estimated distance to **G**.
4. Repeat until **G** is reached.

Result

The algorithm reaches the destination quickly by always choosing the most promising node.

Interpretation

Greedy Best-First Search is faster than many search methods but does **not always guarantee the minimum-cost path**.